Kush Patel

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SUMMARY

Data Science graduate student at Illinois Institute of Technology with experience in building scalable machine learning systems and data-driven analytics. Passionate about applying computational modeling and data engineering to advance precision diverse domain insight generation. Skilled in developing reproducible, interpretable, and cloud-deployed ML solutions integrating datasets to improve business outcomes.

SKILLS

Programming Languages : Python, R, SQL, C

Data Analysis Tools/ Libraries : Pandas, Numpy, Scikit Learn, Matplotlib, Excel, Tableau, Power Bi

Big Data Technologies : Hadoop, Spark

Machine Learning/AI Libraries/Frameworks :RAG, NLP, Google AI Studio, PyTorch, TensorFlow, Keras

EXPERIENCE

Machine Learning Intern

Internshipstudio **May 2023 - June 2023, Remote, India**

• Built and optimized regression and classification models in Python to forecast engagement and ROI metrics.
• Developed automated data preprocessing and feature validation pipelines, reducing manual effort by 20%.
• Designed and evaluated model calibration and interpretability reports to improve business reliability.

Data Analyst Intern

Trainity **Nov 2022 - Dec 2022, Remote, India**

• Conducted exploratory data analysis and statistical modeling on large-scale datasets using SQL and Python.
• Designed KPI-driven dashboards in Power BI to visualize performance and user-behavior metrics.
• Collaborated with cross-functional teams to translate analytical insights into strategic business actions.

ARC Math Tutor

Illinois Institute of Technology **November 2025 - Present, Chicago, IL**

• Mentoring undergraduate and graduate students in mathematics, including Calculus, Linear Algebra, and Probability Theory.
• Translating complex quantitative concepts into actionable learning strategies, improving student comprehension and problem-solving skills in high-level coursework.
• Collaborating with the Academic Resource Center to maintain academic excellence and facilitate personalized support for diverse learning styles.

PROJECT

CareBot: Clinical-Support Medical Chatbot

May 2025 - November 2025

• Built a medical chatbot using LangChain, Pinecone vector search, and Gemini LLMs, delivering 40% faster symptom analysis responses.
• Engineered a PDF-based RAG pipeline enabling 30% more accurate retrieval of medical guidelines through context-aware vector search.
• Enabled clinicians to access summarized insights faster than manual document review, improving triage decision making efficiency.

Diabetes Prediction and Risk Stratification

June 2025 - August 2025

• Designed a predictive framework integrating CDC BRFSS and Kaggle datasets to identify key diabetes risk factors.
• Applied SQL-based feature engineering and interpretable ML models (Logistic Regression, XGBoost, SHAP) for insight generation.
• Created Power BI dashboards to visualize population-level diabetes risk and intervention priorities.

Biological Age Prediction from DNA Methylation (Whole Blood)

March 2025 - May 2025

• Developed an end-to-end, reproducible ML pipeline to estimate chronological age from DNA methylation (DNAm) arrays, often used as a proxy for biological aging.
• Implemented interpretable baseline models (Ridge, PCR, PLS, Random Forest) and a light stacked ensemble for improved accuracy and calibration.
• Built robust data alignment and validation workflows to ensure cross-study generalization on an external cohort (GSE157131).

Dashask: Auto-Grading Subjective Test Platform

February 2021 - April 2021

• Built an NLP-driven evaluation engine using BERT embeddings and cosine similarity for automated subjective grading.
• Conducted reliability testing and calibration improvements, achieving a 15% gain in scoring consistency.
• Integrated troubleshooting and error-handling pipelines, reducing manual grading workload by more than 65%.

EDUCATION

Master of Applied Science (Data Science)

Illinois Institute of Technology • Chicago, IL • May 2026 • 3.77

• Event Coordinator in Indian Student Association

• Academic Resource Center Math Tutor

B.Tech in Computer Science and Engineering

University of Mumbai • India • May 2023 • 8.6

PUBLICATIONS

Automatic Subjective Answer Grading Software

Professional • IEEE • 2023

• Built an ML + NLP-based subjective answer grading system achieving 85% grading accuracy by computing semantic similarity between student and ideal responses.

• Developed a Django web application supporting 100+ concurrent users with <2s grading latency, enabling near real-time evaluation during online exams.

• Reduced manual grading effort by 60–70% using a human-in-the-loop workflow, allowing admins to review and override automated scores.